

**Big Data  
Laporan HDFS**



**Disusun Oleh:**

Much. Zannur Nadhif      TI-3A/2341720260

**PROGRAM STUDI D-IV TEKNIK  
INFORMATIKA JURUSAN TEKNOLOGI  
INFORMASI POLITEKNIK NEGERI  
MALANG**

HadoopOverviewDatanodesDatanode Volume FailuresSnapshotStartup ProgressUtilities

Overview

'hadoop-namenode:9000' (✓active)

|                |                                                             |
|----------------|-------------------------------------------------------------|
| Started:       | Thu Apr 30 13:54:25 +0700 2026                              |
| Version:       | 3.4.1, #4d7825309348956336b8f06a08322b78422849b1            |
| Compiled:      | Wed Oct 09 21:57:00 +0700 2024 by mthakur from branch-3.4.1 |
| Cluster ID:    | CID-b2f85194-b9ae-4e40-b695-f5db19fb31b9                    |
| Block Pool ID: | BP-1679859687-192.168.1.15-1747046626325                    |

Summary

Security is off.  
SafeMode is off.  
1 files and directories, 0 blocks (0 replicated blocks, 0 erasure coded block groups) = 1 total filesystem object(s).  
Heap Memory used 49.91 MB of 110 MB Heap Memory. Max Heap Memory is 492 MB.  
Non Heap Memory used 99.71 MB of 103.09 MB Committed Non Heap Memory. Max Non Heap Memory is <unbounded>.

|                                                        |                                          |
|--------------------------------------------------------|------------------------------------------|
| Configured Capacity:                                   | 329.04 GB                                |
| Configured Remote Capacity:                            | 0 B                                      |
| DFS Used:                                              | 96 KB (0%)                               |
| Non DFS Used:                                          | 20.36 GB                                 |
| DFS Remaining:                                         | 291.84 GB (88.69%)                       |
| Block Pool Used:                                       | 96 KB (0%)                               |
| DataNodes usages% (Min/Median/Max/stdDev):             | 0.00% / 0.00% / 0.00% / 0.00%            |
| Live Nodes                                             | 3 (Decommissioned: 0, In Maintenance: 0) |
| Dead Nodes                                             | 0 (Decommissioned: 0, In Maintenance: 0) |
| Decommissioning Nodes                                  | 0                                        |
| Entering Maintenance Nodes                             | 0                                        |
| Total Datanode Volume Failures                         | 0 (0 B)                                  |
| Number of Under-Replicated Blocks                      | 0                                        |
| Number of Blocks Pending Deletion (including replicas) | 0                                        |
| Block Deletion Start Time                              | Thu Apr 30 13:54:25 +0700 2026           |
| Last Checkpoint Time                                   | Tue May 05 04:56:52 +0700 2026           |
| Last HA Transition Time                                | Never                                    |
| Enabled Erasure Coding Policies                        | RS-6-3-1024k                             |

Summary : : Menampilkan total kapasitas memori yang digunakan dan yang tersisa di seluruh server

NameNode Journal Status

Current transaction ID: 260

| Journal Manager                                          | State                                                                                                |
|----------------------------------------------------------|------------------------------------------------------------------------------------------------------|
| FileJournalManager(root=/usr/local/hadoop/data/namenode) | EditLogFileOutputStream(/usr/local/hadoop/data/namenode/current/edits_inprogress_000000000000000260) |

NameNode Storage

| Storage Directory               | Type            | State  |
|---------------------------------|-----------------|--------|
| /usr/local/hadoop/data/namenode | IMAGE_AND_EDITS | Active |

DFS Storage Types

| Storage Type | Configured Capacity | Capacity Used | Capacity Remaining | Block Pool Used | Nodes In Service |
|--------------|---------------------|---------------|--------------------|-----------------|------------------|
| DISK         | 329.04 GB           | 96 KB (0%)    | 291.84 GB (88.69%) | 96 KB           | 3                |

Hadoop, 2024.

NameNode Journal Status: Menunjukkan apakah catatan transaksi data berjalan dengan normal

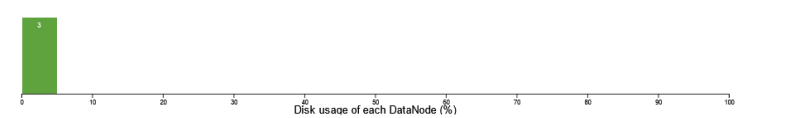
NameNode Storage: Menampilkan lokasi penyimpanan data identitas file di server pusat

DFS Storage types : Menampilkan informasi penggunaan kapasitas berdasarkan tipe media, yang bertujuan agar sistem dapat mengatur data mana yang perlu diproses cepat dan data mana yang cukup disimpan dalam gudang arsip

Datanode Information

✔ In service
❌ Down
🔄 Decommissioning
🚫 Decommissioned
⛔ Decommissioned & dead
  
🛠 Entering Maintenance
🔧 In Maintenance
🛑 In Maintenance & dead

Datanode usage histogram



In operation

Datanode State: All
Show: 25 entries
Search:

| Node                                                  | Http Address                                                              | Last contact | Last Block Report | Used  | Non DFS Used | Capacity  | Blocks | Block pool used | Block pool usage StdDev | Version |
|-------------------------------------------------------|---------------------------------------------------------------------------|--------------|-------------------|-------|--------------|-----------|--------|-----------------|-------------------------|---------|
| ✔/default-hadoop-datanode1-9896 (192.168.57.154-9896) | <a href="http://hadoop-datanode1-9896/">http://hadoop-datanode1-9896/</a> | 2s           | 230m              | 32 KB | 6.78 GB      | 100.68 GB | 0      | 32 KB (0%)      | 0%                      | 3.4.1   |
| ✔/default-hadoop-datanode2-9896 (192.168.57.208-9896) | <a href="http://hadoop-datanode2-9896/">http://hadoop-datanode2-9896/</a> | 0s           | 317m              | 32 KB | 6.70 GB      | 100.68 GB | 0      | 32 KB (0%)      | 0%                      | 3.4.1   |
| ✔/default-hadoop-datanode3-9896 (192.168.57.155-9896) | <a href="http://hadoop-datanode3-9896/">http://hadoop-datanode3-9896/</a> | 1s           | 89m               | 32 KB | 6.70 GB      | 100.68 GB | 0      | 32 KB (0%)      | 0%                      | 3.4.1   |

Showing 1 to 3 of 3 entries

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Di halaman Datanodes terdapat grafik yang memberikan gambaran visual tentang distribusi penggunaan ruang simpan di seluruh serve serta berisi rincian teknis dari server penyimpanan data yang aktif. Kolom Node mengidentifikasi nama host dan alamat IP setiap server. Http Address berisi tautan untuk mengelola server secara individu. Last Contact dan Last Block Report memantau waktu respons serta laporan data terbaru.

Entering Maintenance

No nodes are entering maintenance.

Decommissioning

No nodes are decommissioning.

Hadoop, 2024.

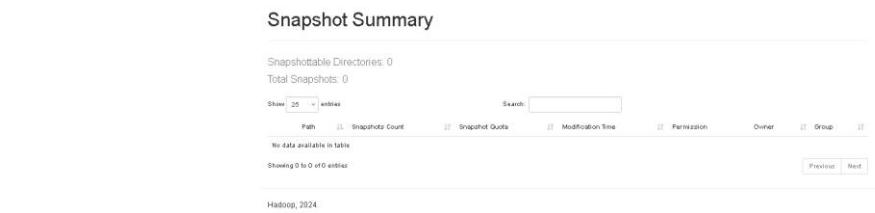
Entering Maintenance menampilkan server yang masuk ke tahap pemeliharaan atau perbaikan teknis. Sedangkan Decommissioning mencatat daftar server yang akan keluar dari jaringan secara permanen. Sistem memindahkan seluruh data ke server lain secara otomatis untuk mencegah kehilangan informasi sebelum mematikan server tersebut.

Datanode Volume Failures

There are no reported volume failures.

Hadoop, 2024.

Seperti namanya ini untuk melaporkan kerusakan atau kegagalan pada media penyimpanan fisik yang digunakan oleh server penyimpan data.



Untuk bagian snapshot sendiri memiliki fungsi sebagai Halaman Snapshot Summary mencatat daftar folder yang dapat user buat kan salinan datanya secara instan. User menggunakan fitur ini untuk memulihkan data jika terjadi kesalahan atau kehilangan file pada waktu tertentu

Startup Progress

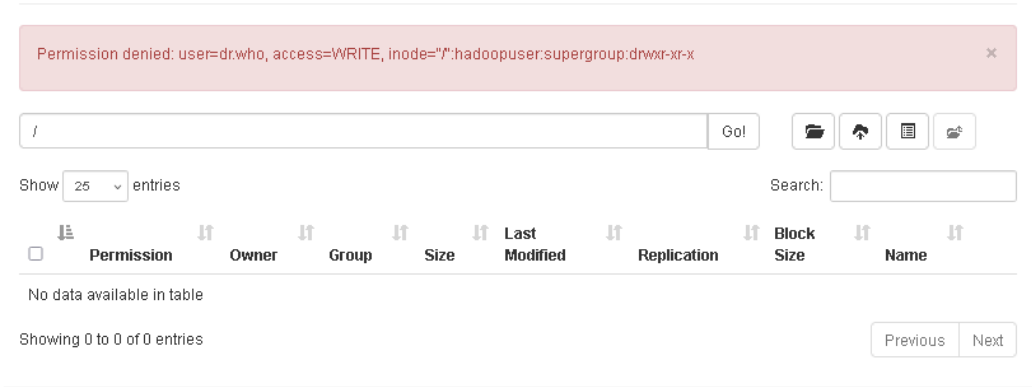
Elapsed Time: 0 sec, Percent Complete: 100%

| Phase                                                                                                  | Completion | Elapsed Time |
|--------------------------------------------------------------------------------------------------------|------------|--------------|
| <b>Loading fsimage</b> /usr/local/hadoop/data/hamenode/current/fsimage_000000000000000034 405 B        | 100%       | 0 sec        |
| erasure coding policies (0/0)                                                                          | 100%       |              |
| inodes (1/1)                                                                                           | 100%       |              |
| delegation tokens (0/0)                                                                                | 100%       |              |
| cache pools (0/0)                                                                                      | 100%       |              |
| <b>Loading edits</b>                                                                                   | 100%       | 0 sec        |
| /usr/local/hadoop/data/hamenode/current/edits_000000000000000035-000000000000000035 1 MB (1/1)         | 100%       |              |
| <b>Saving checkpoint</b>                                                                               | 100%       | 0 sec        |
| erasure coding policies /usr/local/hadoop/data/hamenode/current/fsimage.chkpt_000000000000000035 (0/0) | 100%       |              |
| inodes /usr/local/hadoop/data/hamenode/current/fsimage.chkpt_000000000000000035 (0/0)                  | 100%       |              |
| delegation tokens /usr/local/hadoop/data/hamenode/current/fsimage.chkpt_000000000000000035 (0/0)       | 100%       |              |
| cache pools /usr/local/hadoop/data/hamenode/current/fsimage.chkpt_000000000000000035 (0/0)             | 100%       |              |
| <b>Safe mode</b>                                                                                       | 100%       | 0 sec        |
| awaiting reported blocks (0/0)                                                                         | 100%       |              |

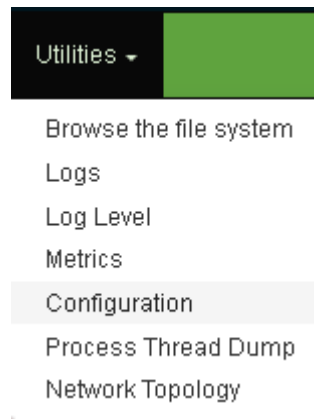
Hadoop, 2024.

Sebagai penanda akan tahapan proses pemuatan sistem HDFS saat pertama kali dijalank

Browse Directory



Berfungsi sebagai pengelola file berbasis web untuk menjelajahi isi direktori dalam sistem HDFS



- Logs. Bagian ini berisi berkas catatan sistem
- Log Level. Fitur ini memungkinkan User mengubah tingkat detail laporan aktivitas.
- Metrics. User bisa melihat statistik performa klaster secara langsung. Data ini membantu User memantau efisiensi kerja seluruh sistem.
- Configuration. Halaman ini menampilkan semua daftar pengaturan yang sedang aktif.
- Process Thread Dump. User mendapatkan informasi tentang tugas yang sedang diproses oleh sistem
- Network Topology. Bagian ini memperlihatkan susunan jaringan antar server. User bisa mengetahui posisi server dalam rak jaringan melalui peta ini.